

YEAR 7 • UNIT 1 • EXTRA HOMEWORK

Bring Down the Next Digit

Two-digit multiplication • and long division, written out

Name: _____

Class: _____

Date given: _____

Date due: _____

Why this sheet exists

Every method in Unit 1 finishes in one of **two** operations. You find a factor by **dividing**. You find the HCF by dividing twice. You check a square past 12×12 by **multiplying**, and a cube by multiplying twice.

Last time you divided by counting up in multiples. That works until the number gets big. This sheet is the **written method** — the one that does not fall over on 4 digits.

No calculator, and show your working. Working you can see is working you can check.

Part 1 • The facts both methods run on

E1 Fill in the missing number

Every long multiplication and every long division is these facts, one after another. Aim to know them without counting.

(a) $7 \times 6 =$ _____ (e) $9 \times 8 =$ _____ (i) $9 \times 7 =$ _____ (m) $12 \times 7 =$ _____

(b) $8 \times 7 =$ _____ (f) $7 \times 7 =$ _____ (j) $6 \times 6 =$ _____ (n) $11 \times 8 =$ _____

(c) $9 \times 6 =$ _____ (g) $4 \times 9 =$ _____ (k) $9 \times 9 =$ _____ (o) $12 \times 9 =$ _____

(d) $6 \times 8 =$ _____ (h) $8 \times 8 =$ _____ (l) $8 \times 4 =$ _____ (p) $7 \times 4 =$ _____

E2 Multiply by a ten

The second row is always a ten

To multiply by **20**: multiply by **2**, then **put a zero on the end**. $34 \times 2 = 68$, so $34 \times 20 = 680$. That is the whole trick, and it is what the second row of a long multiplication is.

(a) $26 \times 10 =$ _____ (c) $47 \times 20 =$ _____ (e) $18 \times 60 =$ _____

(b) $26 \times 30 =$ _____ (d) $52 \times 40 =$ _____ (f) $73 \times 50 =$ _____

E3 Count up in a two-digit number**Do this before you divide**

Before dividing by 23, write out the first few multiples of 23. Then the division is only **reading your own list**. This is two minutes that saves ten.

×	1	2	3	4	5	6
23						
16						
24						
37						

Part 2 · Two rows, then add**The method, once, in full**

$$\begin{array}{r}
 58 \\
 \times 34 \\
 \hline
 232 \\
 1740 \\
 \hline
 1972
 \end{array}$$

Row 1: $58 \times 4 = 232$.**Row 2:** $58 \times 30 = 1740$ – multiply by 3, put the zero on.**Add them:** $232 + 1740 = 1972$.

One row means you forgot the tens. Your answer comes out about ten times too small. **Two rows, every time.**

E4 Warm up: two digits by one digit

53×7	68×6	87×9	76×8
94×4	47×8	65×7	89×6
58×9	73×6	96×7	84×5

E5 Two digits by two digits**Two rows, then add.** Show both rows every time.

32×14	45×23	28×36
57×42	63×27	86×35
74×58	49×61	95×24
38×47	84×73	66×52

E6 Three digits by two digits**Nothing new here**

Three digits by two is still **two rows**. Row 1 is the whole number times the units. Row 2 is the whole number times the tens. It is only longer, not harder.

124×32	236×18	407×25
318×46	529×37	324×18

Look back. The last one, 324×18 , is the second half of a cube root: $18 \times 18 = 324$, and then 324×18 . So $\sqrt[3]{\quad} = 18$.

Part 3 · Long division: bring down the next digit

Word help – the words in a division

divisor	the number you are dividing by . In $966 \div 23$ it is 23.
goes into	"23 goes into 96 four times" means $96 \div 23 = 4$, with some left over.
bring down	take the next digit of the big number and write it beside what is left.
remainder	what is left at the very end. We write $75 \div 8 = 9 \text{ r } 3$.

Divide, multiply, subtract, bring down – then start again

$$\begin{array}{r}
 42 \\
 23 \overline{) 966} \\
 \underline{92} \\
 46 \\
 \underline{46} \\
 0
 \end{array}$$

Divide: 23 goes into 96 **four** times. Write the 4 on top, above the 6.

Multiply: $4 \times 23 = 92$. Write it underneath.

Subtract: $96 - 92 = 4$.

Bring down the next digit, the 6, to make **46**.

Now start again: 23 goes into 46 **twice**, $2 \times 23 = 46$, $46 - 46 = 0$. Nothing is left, so $966 \div 23 = 42$.

E7 Warm up: one-digit divisors, no remainder

- | | |
|--------------------------|---------------------------|
| (a) $252 \div 6 =$ _____ | (e) $1344 \div 4 =$ _____ |
| (b) $469 \div 7 =$ _____ | (f) $2016 \div 6 =$ _____ |
| (c) $584 \div 8 =$ _____ | (g) $3528 \div 7 =$ _____ |
| (d) $765 \div 9 =$ _____ | (h) $5832 \div 8 =$ _____ |

E8 One-digit divisors, with a remainder

Write each answer as a whole number and a remainder, like this: $75 \div 8 = 9 \text{ r } 3$.

- | | |
|--------------------------|---------------------------|
| (a) $437 \div 6 =$ _____ | (d) $1000 \div 7 =$ _____ |
| (b) $529 \div 7 =$ _____ | (e) $2345 \div 8 =$ _____ |
| (c) $815 \div 9 =$ _____ | (f) $4567 \div 9 =$ _____ |

E9 Two-digit divisors – set it out properly

These all come out exactly. Show the full method in each box.

Write the list first

Before you start, write the first six multiples of the divisor in the corner of the box. Then every “how many times does it go in” is **reading**, not guessing.

$768 \div 16$

$851 \div 23$

$1224 \div 24$

$2072 \div 37$

$4212 \div 18$

$3375 \div 15$

$1088 \div 32$

$1222 \div 47$

$986 \div 29$

$2408 \div 43$

$1872 \div 26$

$2940 \div 35$

E10 Two-digit divisors, with a remainder

The remainder is always **smaller than the divisor**. Left with 30 after dividing by 24? Then 24 goes in **one more time** — go back.

$1000 \div 24$	$875 \div 32$
$2500 \div 36$	$1500 \div 45$
$3000 \div 28$	$1800 \div 26$

E11 Find the missing number

Every multiplication is two divisions

If $24 \times 51 = 1224$, then $1224 \div 24 = 51$ **and** $1224 \div 51 = 24$. Look at **which** number is missing before you decide what to do: if the **big** number is missing, you **multiply**.

(a) $24 \times \underline{\hspace{2cm}} = 1224$

(i) $43 \times \underline{\hspace{2cm}} = 2408$

(b) $\underline{\hspace{2cm}} \times 16 = 768$

(j) $\underline{\hspace{2cm}} \times 26 = 1872$

(c) $2072 \div \underline{\hspace{2cm}} = 56$

(k) $2940 \div \underline{\hspace{2cm}} = 84$

(d) $\underline{\hspace{2cm}} \div 18 = 234$

(l) $\underline{\hspace{2cm}} \div 29 = 34$

(e) $37 \times \underline{\hspace{2cm}} = 851$

(m) $47 \times \underline{\hspace{2cm}} = 1222$

(f) $\underline{\hspace{2cm}} \times 18 = 324$

(n) $\underline{\hspace{2cm}} \times 32 = 1088$

(g) $3375 \div \underline{\hspace{2cm}} = 225$

(o) $4212 \div \underline{\hspace{2cm}} = 234$

(h) $\underline{\hspace{2cm}} \div 8 = 729$

(p) $\underline{\hspace{2cm}} \div 15 = 225$

E12 Mixed drill — no boxes, no help

Both operations, mixed up. Work down the columns as fast as you can be accurate. **Two pairs of these have the same answer — find both pairs.**

- | | | |
|----------------------------|----------------------------|----------------------------|
| 1. $480 \div 6 =$ _____ | 11. $936 \div 12 =$ _____ | 21. $999 \div 3 =$ _____ |
| 2. $24 \times 30 =$ _____ | 12. $18 \times 15 =$ _____ | 22. $44 \times 25 =$ _____ |
| 3. $3600 \div 9 =$ _____ | 13. $2400 \div 16 =$ _____ | 23. $864 \div 12 =$ _____ |
| 4. $45 \times 12 =$ _____ | 14. $27 \times 40 =$ _____ | 24. $125 \times 8 =$ _____ |
| 5. $720 \div 8 =$ _____ | 15. $546 \div 7 =$ _____ | 25. $756 \div 7 =$ _____ |
| 6. $16 \times 25 =$ _____ | 16. $23 \times 30 =$ _____ | 26. $48 \times 25 =$ _____ |
| 7. $1000 \div 8 =$ _____ | 17. $1250 \div 25 =$ _____ | 27. $630 \div 15 =$ _____ |
| 8. $60 \times 14 =$ _____ | 18. $19 \times 21 =$ _____ | 28. $26 \times 40 =$ _____ |
| 9. $812 \div 4 =$ _____ | 19. $4000 \div 25 =$ _____ | 29. $1440 \div 12 =$ _____ |
| 10. $35 \times 11 =$ _____ | 20. $32 \times 15 =$ _____ | 30. $55 \times 12 =$ _____ |

Part 4 • Use it**Word help — which operation does the sentence want?**

each / every — usually multiply. 32 boxes with 124 in **each** is 32×124 .

in rows of • shared equally between — divide. **how many are left over** — the remainder.

Write the calculation first, then the answer.

1. **The hall.** There are **1224 chairs** in the hall, set out in rows of **24**. How many rows are there?

2. **The books.** Mr Bowen orders **32 boxes** of books with **124 books** in each box. How many books is that altogether?

3. **The pencils.** Mr Bowen has **1000 pencils** and shares them equally between **24 classes**. How many does each class get, and how many are left over?

4. **The order.** The school orders **18 packs** of pens with **236 pens** in each pack. The pens are then shared equally between **24 classrooms**. How many pens does each classroom get?

5. **The tiles.** Mr Bowen tiles a floor with **47 rows** of **26 tiles**. The tiles come in boxes of **20**. How many tiles does he need, and how many boxes must he buy?

6. **The stickers.** Mr Bowen has **4212 stickers** and sticks **18** of them on every page of a sticker book. Each book has **50 pages**. How many pages does he fill, and how many **whole books** is that?

Answer Key – Teacher Copy

E1 Fill in the missing number

- (a) 42 (b) 56 (c) 54 (d) 48 (e) 72 (f) 49 (g) 36 (h) 64
 (i) 63 (j) 36 (k) 81 (l) 32 (m) 84 (n) 88 (o) 108 (p) 28

Time this one rather than marking it. Under two minutes for all sixteen and the tables are not what is slowing them down — look at Part 3 instead. The three worth re-drilling are (b) 8×7 , (e) 9×8 and (i) 9×7 : they turn up in both rows of a long multiplication *and* in every long division by 7, 8 or 9, so an error there is paid for three times.

E2 Multiply by a ten

- (a) 260 (b) 780 (c) 940 (d) 2080 (e) 1080 (f) 3650

This is the second row of E5 taken out and practised alone, which is why it comes before the method. The error to look for is a **missing zero** — 78 instead of 780 in (b). A student who makes it here makes it in every question in E5, and it is far faster to catch on this line than inside a four-line sum.

E3 Count up in a two-digit number

- 23 23, 46, 69, 92, 115, 138
 16 16, 32, 48, 64, 80, 96
 24 24, 48, 72, 96, 120, 144
 37 37, 74, 111, 148, 185, 222

This question is the whole sheet in miniature and it is the one to insist on. A student who writes the list before dividing is *reading* at every step of a long division; a student who does not is guessing, rubbing out and guessing again. Note that these four are exactly the divisors in E9 — 23 and 16 and 24 — so a student who does this table properly has already done half of E9.

E4 Warm up: two digits by one digit

- $53 \times 7 = 371$ $68 \times 6 = 408$ $87 \times 9 = 783$ $76 \times 8 = 608$
 $94 \times 4 = 376$ $47 \times 8 = 376$ $65 \times 7 = 455$ $89 \times 6 = 534$
 $58 \times 9 = 522$ $73 \times 6 = 438$ $96 \times 7 = 672$ $84 \times 5 = 420$

The error here is carrying, not the tables. 87×9 carries at both columns and is the one to check. 94×4 and 47×8 both come to **376**, and that is not a coincidence: halving one number and doubling the other leaves the answer alone. A student who spots it has found a genuinely useful shortcut — 16×25 in E12 is the same trick, and becomes 8×50 , then 4×100 .

E5 Two digits by two digits

- | | | |
|-----------------------|-----------------------|-----------------------|
| $32 \times 14 = 448$ | $45 \times 23 = 1035$ | $28 \times 36 = 1008$ |
| $57 \times 42 = 2394$ | $63 \times 27 = 1701$ | $86 \times 35 = 3010$ |
| $74 \times 58 = 4292$ | $49 \times 61 = 2989$ | $95 \times 24 = 2280$ |
| $38 \times 47 = 1786$ | $84 \times 73 = 6132$ | $66 \times 52 = 3432$ |

Mark the *working*, not just the answer. **One row of working is the whole diagnosis:** the student multiplied by the units digit and stopped, so the answer is roughly a tenth of the right size — 448 becomes 128. That is a method error and needs re-teaching. An answer out by a small amount with two rows present is arithmetic, and needs a re-check, not a re-teach.

E6 Three digits by two digits

$$124 \times 32 = 3968 \quad 236 \times 18 = 4248 \quad 407 \times 25 = 10175$$

$$318 \times 46 = 14628 \quad 529 \times 37 = 19573 \quad 324 \times 18 = 5832$$

Look back: $\sqrt[3]{5832} = 18$.

407×25 is the one with the **zero in the middle** and it is the interesting failure: students skip the zero column and shorten the row by a digit.

324×18 is not decoration. $18 \times 18 = 324$, so this multiplication is the second half of proving a cube root by hand, and a student who reaches 5832 has done something worth saying out loud. It also reappears as E7(h), $5832 \div 8$, from the other direction.

E7 One-digit divisors, no remainder

(a) 42 (b) 67 (c) 73 (d) 85 (e) 336 (f) 336 (g) 504 (h) 729

(e) and (f) both come to 336 — if a student notices, that is a good sign and worth a word. (h) $5832 \div 8 = 729$ is the same number as E6's last multiplication seen from the other end, and 729 is 9^3 ; a class that spots that has understood something about Unit 1 that no exercise can be set for.

E8 One-digit divisors, with a remainder

(a) 72 r 5 (b) 75 r 4 (c) 90 r 5 (d) 142 r 6 (e) 293 r 1 (f) 507 r 4

The diagnosis is in the remainder. A remainder **bigger than the divisor** is a method error — the student stopped dividing too early — and needs re-teaching. A remainder wrong by one or two is arithmetic and needs a re-check. Only the first is worth class time.

(c) and (f) are the two with a **zero in the answer**: 815 divided by 9 gives 9, then nothing, then 0 — and the zero must be written. A student who answers "9 r 5" has dropped a digit, and that is the single most common long-division error there is.

E9 Two-digit divisors — set it out properly

$$768 \div 16 = 48 \quad 851 \div 23 = 37 \quad 1224 \div 24 = 51$$

$$2072 \div 37 = 56 \quad 4212 \div 18 = 234$$

$$3375 \div 15 = 225 \quad 1088 \div 32 = 34$$

$$1222 \div 47 = 26 \quad 986 \div 29 = 34$$

$$2408 \div 43 = 56 \quad 1872 \div 26 = 72 \quad 2940 \div 35 = 84$$

Look in the corner of the box before you look at the answer. A student who wrote the multiples list first and got it wrong has made an arithmetic slip; a student with no list who got it wrong has been guessing, and telling them the answer teaches them nothing.

$1224 \div 24$ has a **zero step** in the middle of the method — 24 does not go into 12, so you take 122 — and it is the one that stalls people. $3375 \div 15$ is the hardest, and 225 is 15^2 , which is the patio from the lesson.

E10 Two-digit divisors, with a remainder

$$1000 \div 24 = 41 \text{ r } 16 \quad 875 \div 32 = 27 \text{ r } 11 \quad 2500 \div 36 = 69 \text{ r } 16$$

$$1500 \div 45 = 33 \text{ r } 15 \quad 3000 \div 28 = 107 \text{ r } 4 \quad 1800 \div 26 = 69 \text{ r } 6$$

Check the remainders first: any remainder of 24 or more in the first one is a stopped-too-early error, and the student can find it themselves if you ask "can 24 go in again?". That question is worth more than the correction.

$3000 \div 28$ is the hard one, and it is hard for a reason worth naming: 28 does not go into 30 twice, it goes once, and the second digit of the answer is a 0. Expect "17 r 4".

$1000 \div 24$ is the same division as Part 4 question 3, so a student who has done it here has done it there.

E11 Find the missing number

(a) 51 (b) 48 (c) 37 (d) 4212 (e) 23 (f) 18 (g) 15 (h) 5832
 (i) 56 (j) 72 (k) 35 (l) 986 (m) 26 (n) 34 (o) 18 (p) 3375

(d), (h), (l) and (p) are the hard ones, because the missing number is the *big* one and the student has to **multiply** rather than divide: 234×18 , 729×8 , 34×29 , 225×15 . A student who tries to divide there has not looked at which number is missing, and that is a reading error, not an arithmetic one.

Every number in this question appears somewhere else on the sheet, so it is also a check on whether the earlier work was done or copied — a student who did E9 properly can fill in half of this from memory.

E12 Mixed drill — no boxes, no help

1. 80 2. 720 3. 400 4. 540 5. 90 6. 400 7. 125 8. 840 9. 203 10. 385
 11. 78 12. 270 13. 150 14. 1080 15. 78 16. 690 17. 50 18. 399 19. 160
 20. 480
 21. 333 22. 1100 23. 72 24. 1000 25. 108 26. 1200 27. 42 28. 1040 29.
 120 30. 660

The two pairs: 3 and 6 both make **400**; 11 and 15 both make **78**.

Time this one and mark it as a block — it is a speed question, not a method question, and the whole diagnosis is *where they slowed down*. A student who is fast on the multiplications and slow on the divisions has the tables but not the division method, which is the opposite way round from what most people assume.

9 ($812 \div 4 = 203$) and 25 ($756 \div 7 = 108$) are the **zero in the answer** again, and 7 ($1000 \div 8 = 125$) is the one they will want a calculator for. The pairs are there to make them look back over their own work, which is the check they never do voluntarily.

E13 Use it

1. $1224 \div 24 = 51$ rows.
2. $32 \times 124 = 3968$ books.
3. $1000 \div 24 = 41 \text{ r } 16$: **41 pencils** each, with **16 left over**.
4. $18 \times 236 = 4248$ pens, then $4248 \div 24 = 177$ pens each.
5. $47 \times 26 = 1222$ tiles. $1222 \div 20 = 61 \text{ r } 2$, so he must buy **62 boxes**.
6. $4212 \div 18 = 234$ pages. Then $234 \div 50 = 4 \text{ r } 34$, so **234 pages**, which is **4 whole books** (with 34 pages of a fifth).

1 and 2 are E9 and E6 wearing a story, so a student who can do those and not these has a reading gap, not an arithmetic one — read the sentence back to them rather than reworking the sum.

3 is the sentence test: “41.67” is a correct division answering a question nobody asked, because you cannot give a class two-thirds of a pencil.

4 is 236×18 from E6 followed by a division, and it comes out *exactly*, which is the reward for getting the multiplication right. A student who reaches 4248 and then a messy remainder has an error in step one, not step two — send them back to the multiplication rather than to the division.

5 is the one to spend time on. $47 \times 26 = 1222$ is E9 seen from the other side, but the second half is the only place on the sheet where the remainder makes the answer go **up**: 61 boxes is not enough, because the last 2 tiles still have to come in a box. Expect “61” and “61 r 2” from most of the class, and treat it as a discussion rather than a cross — the arithmetic is right and the reading is not.

6 is the last thing on the sheet and the only one where the remainder has to be *read* rather than reported: 4 r 34 means four full books and part of a fifth, not “4.68 books”. The common failure is stopping at 234 and calling those books.